

JOAO PESSOA, Brazil

WMO: 827980

Lat: 7.100S

Lon: 34.867W

Elev: 7

StdP: 101.24

Time Zone: -3.00 (W03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.9	20.2	17.1	12.2	27.7	17.9	12.9	26.4	9.6	27.2	9.2	27.3	2.2	210	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.0	32.0	24.6	31.2	24.4	31.1	24.3	26.2	29.5	25.8	29.2	25.6	28.9	4.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	27.5	25.0	20.1	27.4	24.2	19.2	27.0	81.2	29.8	79.8	29.7	78.3	29.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.9	7.3	DB	19.2	32.4	0.8	0.5	18.6	32.7	18.2	33.0	17.7	33.3	17.1	33.6	(4)
(5)			WB	18.1	26.6	0.7	0.8	17.6	27.1	17.2	27.6	16.8	28.1	16.4	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	27.0	27.2	27.4	27.0	26.4	25.4	24.7	24.6	25.2	26.1	26.7	27.0	(6)
(7)		DBStd	1.27	0.79	0.84	0.85	0.89	0.84	0.91	0.77	0.74	0.81	0.75	0.67	0.75	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5919	528	482	539	512	509	461	454	452	456	498	501	528	(10)
(11)		CDD18.3	2878	269	249	281	262	251	211	196	194	206	240	251	269	(11)
(12)		CDH23.3	24375	2614	2488	2828	2414	2010	1362	1068	1157	1422	1990	2370	2652	(12)
(13)		CDH26.7	7158	832	832	966	774	574	320	185	217	333	557	731	838	(13)
(14)	Wind	WSAvg	3.4	3.3	3.2	3.0	2.9	3.1	3.3	3.6	3.8	3.9	3.7	3.5	3.4	(14)
(15)	Precipitation	PrecAvg	971	47	65	104	161	157	173	131	52	32	13	11	23	(15)
(16)		PrecMax	1539	187	163	222	334	361	334	252	136	107	45	27	75	(16)
(17)		PrecMin	491	10	4	27	44	29	35	40	9	9	1	1	6	(17)
(18)		PrecStd	251	37	42	55	81	77	72	60	31	25	10	6	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.2	32.2	32.2	31.2	30.2	29.2	29.2	30.2	31.1	31.2	31.9	(19)
(20)			MCWB	24.0	24.6	24.9	24.8	24.9	24.2	23.1	22.8	23.0	23.1	23.2	23.9	(20)
(21)		2%	DB	31.1	31.8	32.0	31.8	30.9	29.9	29.0	29.0	29.8	30.2	31.0	31.1	(21)
(22)			MCWB	24.1	24.6	24.8	24.9	24.8	24.0	23.1	22.7	22.9	23.0	23.4	23.8	(22)
(23)		5%	DB	30.9	31.0	31.2	31.0	30.1	29.1	28.2	28.2	29.1	30.0	30.2	30.9	(23)
(24)			MCWB	24.1	24.5	24.7	24.8	24.5	23.6	22.9	22.6	22.7	23.0	23.2	23.8	(24)
(25)		10%	DB	30.1	30.2	30.9	30.2	29.8	28.8	27.9	28.0	28.8	29.2	30.0	30.1	(25)
(26)			MCWB	23.9	24.4	24.7	24.7	24.4	23.5	22.8	22.5	22.7	22.9	23.1	23.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	26.4	26.6	26.7	26.3	25.5	24.9	24.7	24.7	24.8	25.2	25.7	(27)
(28)			MCDWB	29.1	29.8	30.0	30.0	29.4	28.5	27.3	27.2	27.8	28.0	28.9	28.9	(28)
(29)		2%	WB	25.2	25.8	26.1	26.2	25.7	24.9	24.4	24.1	24.1	24.2	24.6	25.2	(29)
(30)			MCDWB	28.8	29.3	29.5	29.4	29.0	28.0	26.9	26.9	27.4	27.8	28.6	28.7	(30)
(31)		5%	WB	25.0	25.5	25.7	25.7	25.4	24.5	23.9	23.6	23.7	24.0	24.2	24.8	(31)
(32)			MCDWB	28.7	29.0	29.2	29.0	28.6	27.5	26.6	26.6	27.0	27.8	28.3	28.5	(32)
(33)		10%	WB	24.6	25.1	25.4	25.4	25.0	24.1	23.6	23.3	23.4	23.7	23.9	24.5	(33)
(34)			MCDWB	28.3	28.8	28.9	28.6	28.1	27.1	26.2	26.4	26.8	27.4	28.0	28.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.0	7.0	6.9	6.7	6.7	6.6	7.3	7.4	7.3	7.4	7.1	(35)
(36)			MCDBR	7.5	7.5	7.5	7.7	7.4	7.4	7.3	7.8	7.8	7.7	7.6	7.5	(36)
(37)			MCWBR	2.4	2.5	2.5	2.7	2.9	3.2	3.2	3.3	2.8	2.5	2.4	2.3	(37)
(38)		5% WB	MCDWB	6.8	6.8	6.9	6.8	6.7	6.6	6.3	7.1	7.2	7.2	7.3	6.9	(38)
(39)			MCWBR	2.5	2.7	2.8	2.9	3.0	3.3	3.2	3.5	3.1	2.7	2.6	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.397	0.401	0.400	0.397	0.388	0.390	0.393	0.403	0.424	0.417	0.403	0.411	(40)	
(41)		taud	2.494	2.491	2.499	2.532	2.553	2.538	2.512	2.446	2.365	2.388	2.439	2.433	(41)	
(42)		Ebn at Noon	942	935	922	898	879	861	866	880	887	909	930	925	(42)	
(43)		Edh at Noon	115	116	113	105	98	97	101	113	127	127	121	121	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.12	6.10	6.03	5.42	4.95	4.47	4.66	5.41	5.92	6.34	6.59	6.32	(44)	
(45)		RadStd	0.37	0.29	0.30	0.36	0.35	0.27	0.33	0.23	0.27	0.24	0.22	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (3 neighbors)	-0.47	-0.30	-0.65	-0.95	-0.46	-0.33	N/A	N/A	-218	-218	

Nomenclature: See separate page